

Summer Internship 25-26
On
Advanced UI/UX Design & Full-Stack Web
Application Development

as the partial fulfillment of
Semester - VI study
for the degree of
Bachelor of Science (Information Technology)
(Affiliated to Sarvajanik University)
during the academic year 2025-2026.

Developed By:

Patel Aryan Shaileshbhai

Enrollment No: SR24BSIT138

Company Name:

Pranayuv Technologies Pvt. Ltd.

www.pranayuv.com



Sarvajanik University
Shree Ramkrishna Institute of Computer Education and Applied Sciences

Behind P.T. Science College, M.T.B. College Campus, Athwalines, Surat - 395 001



SARVAJANIK UNIVERSITY
Shree Ramkrishna Institute of Computer
Education and Applied Sciences

Behind P.T. Science College, M.T.B. College Campus, Athwalines,
Surat - 395 001.



CERTIFICATE

DEPARTMENT OF COMPUTER SCIENCE

This is to certify that

Student Name: **Patel Aryan Shaileshbhai** Enrollment No: **SR24BSIT138**

Degree: **B.Sc. (Information Technology)** Semester: **VI (Academic Year 2025–2026)**

has successfully completed his Summer Internship work 25-26 on

Advanced UI/UX Design & Full-Stack Web Application Development

as the partial fulfillment of

Semester - VI study

for the degree of

Bachelor of Science (Information Technology)

(Affiliated to Sarvajanik University)

during the academic year 2025-2026.

Internship Incharge

Dept. of Computer Science

Head of Department (HOD)

Dept. of Computer Science

Date: _____

Place: Surat, Gujarat

University

Examination

Validation:

Enrollment No. : SR24BSIT138

Signature of External Examiners: 1. _____

2. _____

Figure 31.4: Official Certificate of Internship Completion (Unaltered Original Document) — Pranayuv Technologies.



PRANAYUV TECHNOLOGIES PVT LTD
Empowering Lives through Innovation
CIN U62099AP2025PTC119012

CERTIFICATE OF INTERNSHIP

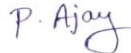
Date: 13-03-2026
Intern Code: PRNV2026011

This is to certify that **Mr. Patel Aryan Shaileshbhai** has successfully completed his internship at Pranayuv Technologies Pvt. Ltd. as a Brand and Growth Intern.

He has successfully completed **180 hours** of internship training as required for his academic curriculum. During this period, he was involved in brand development activities, market research, business growth initiatives, content strategy, campaign support, and related organizational tasks.

Mr. Patel Aryan Shaileshbhai demonstrated professionalism, dedication, and a strong willingness to learn. He completed his assigned responsibilities sincerely and efficiently.

We appreciate his contributions during the internship period and wish him success in his future academic and professional endeavors.


Ajay Polavarapu,
Director,
Pranayuv Technologies Pvt. Ltd.



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 +91 9676448550

 info@pranayuv.com

The following full-page document represents the official corporate Certificate of Internship Completion (Intern Code: PRNV2026011) executed by Director Ajay Polavarapu, confirming successful completion of 180 hours of industrial training.

Internship Completion Certificate

Figure 31.3: Official Internship Offer Letter & NDA (Unaltered Original Document) — Page 3 of 3**PRANAYUV TECHNOLOGIES PVT LTD**

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5. Duration

This Agreement remains in effect during the internship and for a period of two (2) years following its termination, unless otherwise required by law.

6. Non-Compete and Non-Solicitation

This Agreement does not restrict the Intern from working for other entities, provided no confidential Information is used or disclosed in violation of this Agreement. The Intern agrees not to solicit the Company's employees or clients for a period of one (1) year after the internship ends.

7. Miscellaneous

- a. This Agreement constitutes the entire understanding between the Parties regarding the subject matter.
- b. Any amendments must be in writing and signed by both Parties.
- c. If any provision is found invalid, the remaining provisions remain enforceable.

IN WITNESS WHEREOF, the Parties have executed this Agreement as of the Effective Date.

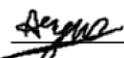
Company

Signature: P. Ajay

Name: p.Ajay

Date: 13/03/26

Intern

Signature: Name: Patel Aryan Shaileshbhai

Date: 13/03/26

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Figure 31.2: Official Internship Offer Letter & NDA (Unaltered Original Document) — Page 2 of 3

NON-DISCLOSURE AGREEMENT

This Non-Disclosure Agreement (the "Agreement") is entered into on 28-02-2026 (the "Effective Date") between Pranayuv Technologies Pvt Ltd ("Company") **Patel Aryan Shaileshbhai** ("Intern"), collectively referred to as the "Parties."

1. Purpose

The Intern will perform services for the Company as part of an internship program. During this engagement, the Intern may have access to or create confidential information. This Agreement ensures the protection of such information.

2. Definition of Confidential Information

"Confidential Information" includes all non-public information disclosed to or created by the Intern during the internship, including but not limited to business plans, financial data, customer lists, software, designs, processes, strategies, and any other proprietary information, whether oral, written, electronic, or otherwise.

3. Obligations of the Intern

The Intern agrees to:

- a. Maintain the confidentiality of all confidential Information and not disclose it to any third party without prior written consent from the Company.
- b. Use confidential Information solely to perform duties assigned by the Company during the internship.
- c. Return or destroy all materials containing Confidential Information upon termination of the internship or at the Company's request.

4. Exclusions from Confidential Information

Confidential Information does not include information that:

- a. Is or becomes publicly available through no fault of the Intern.
- b. Was lawfully in the Intern's possession prior to disclosure by the Company.
- c. Is lawfully obtained from a third party without restriction on disclosure.

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Figure 31.1: Official Internship Offer Letter & NDA (Unaltered Original Document) — Page 1 of 3

**PRANAYUV TECHNOLOGIES PVT LTD**

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INTERNSHIP OFFER LETTERDear Mr. **Patel Aryan Shaileshbhai**,

Welcome to Pranayuv Technologies! We are delighted to have you join our team and look forward to a long and mutually rewarding association.

We are pleased to offer you the position of UI Designer - at Pranayuv Technologies Pvt Ltd ("Pranayuv" or the "Company"). Your date of appointment is effective from your date of joining, which shall be on March 13, 2026. This Internship will be for a period of 6 months, and then you will be selected for full-time based on performance.

Internship Details:**Role:** UI Designer**Department:** Brand & Growth Team**Start Date:** March 13, 2026**Location:** Depending upon the management requirement**Compensation:** No compensation. Upon successful completion, you will receive an experience letter and an internship certificate.

This internship is contingent upon your signing the enclosed Non-Disclosure Agreement (NDA) to protect the confidentiality of our proprietary information.

Please confirm your acceptance of this internship offer by signing below and returning a copy of this letter, along with the signed NDA. For any questions, please get in touch with us at careers@pranayuv.com

We look forward to your contribution to Pranayuv Technologies and providing you with a valuable learning experience. Welcome to the team!

Sincerely,

Ajay Polavarapu

Director

Pranayuv Technologies

Acceptance of Offer:

I, **Patel Aryan Shaileshbhai**, accept the terms of the Internship as outlined in this offer letter. Intern Signature with Date 13/03/26

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The following full-page documents capture the officially executed 3-page Internship Offer Letter and Non-Disclosure Agreement (NDA) between Pranayuv Technologies Pvt. Ltd. and Patel Aryan Shaileshbhai, verified and dated March 13, 2026.

Internship Offer Letter & Non-Disclosure Agreement (NDA) Internship Offer Letter & Non-Disclosure Agreement (NDA) Internship Offer Letter & Non-Disclosure Agreement (NDA)

This concluding appendix incorporates the complete, unaltered original official supporting documents, including the signed Internship Offer Letter, proprietary Non-Disclosure Agreement (NDA), and Certificate of Internship Completion from Pranayuv Technologies Pvt. Ltd. In strict accordance with institutional submission standards, these records are inserted directly as high-resolution full-page documents, preserving their original letterhead, official seals, physical signatures, time-stamping, formatting, and page order without retyping or conversion into editable text.

Official Supporting Documents15 Official Supporting Documents. Appendix – Official Supporting Documents

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1.– Acknowledgement

The practical industrial experience gained during this six-month summer internship at Pranayuv Technologies Pvt. Ltd. has been an invaluable cornerstone in enriching my foundational concepts in computer engineering, modern software architecture, and user interface design. I would like to convey my utmost sincere gratitude to everyone whose continuous guidance, structured feedback, and corporate mentorship contributed to the successful completion of this industrial training.

Foremost, I wish to express my deep sense of gratitude to Mr. Ajay Polavarapu, Director at Pranayuv Technologies Pvt. Ltd., for providing me the remarkable opportunity to work within the Brand & Growth Team as a UI Designer and Full-Stack Developer intern. His profound industry vision regarding healthcare technology, autonomous patient care systems, and digital institutional transformation continuously inspired me to uphold professional excellence and strict academic discipline.

I am equally grateful to my technical project mentors and engineering colleagues at Pranayuv Technologies for their guidance during code reviews, database schema optimizations, and UI/UX usability assessments. Their real-time industry advice enabled me to transition from academic, theoretical exercises to designing production-grade React applications, robust Express REST APIs, and scalable MongoDB databases.

Furthermore, I extend my respectful gratitude to the Head of the Department of Computer Science and the distinguished faculty members of Shree Ramkrishna Institute of Computer Education and Applied Sciences, Sarvajanik University, Surat. Their rigorous academic curriculum and foundational programming instruction prepared me to meet complex real-world software engineering challenges with confidence.

2. Personal Details

Personal Detail / Attribute	Student Specification & Profile
Student Name	Patel Aryan Shaileshbhai
Enrollment Number	SR24BSIT138
Academic Degree	Bachelor of Science (Information Technology) – B.Sc. IT
Current Semester	Semester - VI (Academic Year 2025–2026)
Institution	Shree Ramkrishna Institute of Computer Education and Applied Sciences, Sarvajanik University, Athwalines, Surat - 395 001
Primary Specialization	Full-Stack Web Application Development & Advanced UI/UX Engineering

3. Company Details

The corporate entity hosting the six-month industrial internship training is recognized for its commitment to engineering innovation in healthcare technology and educational management platforms:

Corporate Parameter	Company Profile & Specification
Company Name	Pranayuv Technologies Pvt. Ltd. (CIN: U62099AP2025PTC119012)
Operating Division	Brand & Growth Team
Corporate Director	Mr. Ajay Polavarapu (Director, Pranayuv Technologies)
Official Email	info@pranayuv.com / careers@pranayuv.com
Corporate Contact	+91 967644855 / +91 9676448550
Official Website	www.pranayuv.com
Corporate Motto	Empowering Lives through Innovation

Company Overview

Pranayuv Technologies Pvt. Ltd. is an agile, forward-looking healthcare technology and software solutions organization committed to bridging the digital divide across institutional patient care and educational facility management. Founded on the principle of 'Empowering Lives through Innovation', Pranayuv operates at the synthesis of hardware automation, intuitive digital user interfaces, and robust cloud-based business management software.

The modern healthcare ecosystem frequently encounters operational inefficiencies resulting from fragmented product catalogs, traditional manual quotation workflows, and disparate customer lead tracking. Pranayuv Technologies tackles these challenges by architecting unified, highly secure web applications that connect specialized manufacturing facilities—such as clean room

systems and hospital medical furniture producers—with major healthcare institutions across India.

4. Internship Objectives

The six-month industrial training program at Pranayuv Technologies was structured around well-defined engineering milestones designed to challenge foundational knowledge and build industry-standard competencies in full-stack web architectures and UI/UX design systems:

- **Full-Stack Mastery:** To acquire deep functional expertise in full-stack JavaScript frameworks, specifically building reactive single-page applications using React 19 and scalable modular servers with Express 5 and Node.js.
- **Database Architecture:** To design, normalize, and manage NoSQL document database architectures using MongoDB and Mongoose, implementing compound indexing, schema validation, and relational integrity across complex business models.
- **UI/UX Translation:** To master production-grade visual prototyping and interface design using Figma, translating complex user workflows into intuitive, high-fidelity prototypes and pixel-perfect HTML5/CSS3 implementations.
- **Enterprise Security:** To integrate industry-standard web security practices, including JSON Web Token (JWT) session management, bcrypt password hashing, input sanitization via Joi, and role-based route authentication.

5. Internship Responsibilities

Operating within the Brand & Growth Team under the technical direction of Mr. Ajay Polavarapu, my role as UI Designer Intern and Full-Stack Developer involved comprehensive daily responsibilities spanning the complete software development lifecycle (SDLC):

- **UI/UX Design Engineering:** Conducted requirements analysis, created interactive wireframes, and produced high-fidelity Figma prototypes for academic platforms (EduNexus) and medical furniture e-commerce interfaces.
- **Frontend Implementation:** Developed fully responsive, highly optimized frontend applications using React.js, Tailwind CSS, Vite, and custom CSS3 animations, ensuring seamless cross-device compatibility.
- **Backend API Engineering:** Engineered modular Express REST APIs, constructed secure authentication middleware, and implemented automated file-upload pipelines utilizing Multer for local product image processing.

6. Technology Stack

The engineering execution across all four internship projects utilized a modern, resilient technology stack chosen for enterprise performance, modular scalability, and developer ergonomics.

Technology Domain	Framework / Library Name	Implementation Purpose & Codebase Usage
Frontend UI Framework	React.js (v19.2.5)	Core component driven frontend for Sri Srinivasa Clean Rooms web app; utilizing declarative hooks for state and side-effect management.
Build Tool & Bundler	Vite (v8.0.10)	Next-generation fast bundler providing hot module replacement (HMR), tree-shaking, and optimized production building.
Styling & Design	Tailwind CSS v4 & Vanilla CSS3	Utility-first CSS styling in Project 4 alongside hand-crafted custom CSS animations and variables across static landing projects.
Client Routing	React Router DOM (v7.14.2)	Client-side routing managing seamless SPA page navigation, dynamic URL parameters, and Hash-Link anchor scroll behaviors.
Backend Runtime & Server	Node.js & Express.js (v5.2.1)	Asynchronous event-driven server runtime hosting the RESTful API backend, middleware pipelines, and controller logic.
Database & ODM	MongoDB (v7.2.0) & Mongoose (v9.5.0)	NoSQL document database paired with Mongoose object data modeling for structured schema enforcement and full-text searching.

Technology Domain	Framework / Library Name	Implementation Purpose & Codebase Usage
Authentication & Security	JWT (v9.0.3) & BcryptJS (v3.0.3)	Cryptographic token signing for admin session preservation and 10-round salted password hashing for credential security.
Input Validation	Joi (v18.2.3)	Strict middleware schema verification ensuring API payloads conform to exact datatype and length specifications before processing.
File Upload Middleware	Multer (v2.1.1)	Multipart/form-data processing middleware saving uploaded medical furniture product images directly into local disk <code>/uploads/</code> storage.
UI/UX Design & Prototyping	Figma & Chart.js	Collaborative UI layout drafting, component variant building, and responsive chart graphic rendering in academic management dashboards.

7. Software & Tools Used

Throughout the six-month development tenure, professional IDEs, version control systems, API diagnostic suites, and graphic suites were employed to ensure code quality and rapid delivery:

- **Visual Studio Code (VS Code):** Primary visual Integrated Development Environment (IDE) configured with ESLint (v10.0), Prettier, Tailwind IntelliSense, and Git syntax highlighting extensions for error-free development.

- **Figma Collaborative Design Suite:** Used extensively for interface wireframing, high-fidelity UI layout architecture, and establishing interactive prototype flows before writing frontend code.
- **Postman REST Client:** Essential API diagnostic and test automation software used to construct JSON requests, verify protected JWT authorization headers, and debug Express HTTP status responses.
- **Git & GitHub Actions:** Distributed version control system paired with remote GitHub repository hosting for tracking source code commits, managing feature branches, and securing build milestones.
- **Google Chrome DevTools:** In-depth DOM inspection, responsive device simulation, console network tracking, and memory profile tracing during SPA performance troubleshooting.
- **MongoDB Compass & CLI:** GUI visualizer used to inspect live MongoDB document collections, execute ad-hoc aggregation queries, and verify Index structures on Product and Inquiry schemas.
- **Nodemon & Winston Logger:** Development background utility automating hot server restarts upon file modifications, significantly expediting backend debugging cycles.

8. Project 1 – Pranayuv Landing Page Documentation

The Pranayuv Technology Corporate Landing Page (`index.html`) is a visually prestigious static corporate website crafted without external heavy JavaScript UI frameworks, relying solely on hand-crafted HTML5 markup, modern CSS3 variables, and Vanilla JavaScript. The website serves as the corporate gateway for Pranayuv's hospital equipment and autonomous patient care positioning.

Technical Implementation Details

- **Custom JavaScript Cursor System:** Implemented a sophisticated dual-cursor visual rendering mechanism using JavaScript event listeners (`mousemove`). An instantaneous inner dot tracks cursor coordinates directly, while an outer circle follows with a mathematical interpolation delay, expanding and altering background opacity when hovering over clickable action buttons.
- **HTML5 Canvas Animation Engine:** The full-viewport hero banner utilizes an embedded HTML5 ``<canvas>`` element driven by JavaScript `requestAnimationFrame` cycles to generate dynamic, ambient geometric visual structures representing autonomous digital healthcare.
- **Fluid Typography & CSS Variables:** Leveraged modern CSS ``clamp(min, preferred, max)`` responsive typography functions across three custom font families: Cormorant Garamond for high-end serif titles, Outfit for readable sans-serif body paragraphs, and JetBrains Mono for technical statistics descriptors.
- **Interactive Statistics Metric Bar:** Integrated a bottom statistics metrics grid (e.g., highlighting operational metrics, innovation targets, and institutional partnerships) separated by sophisticated sub-pixel borders and dark contrasting background styling.

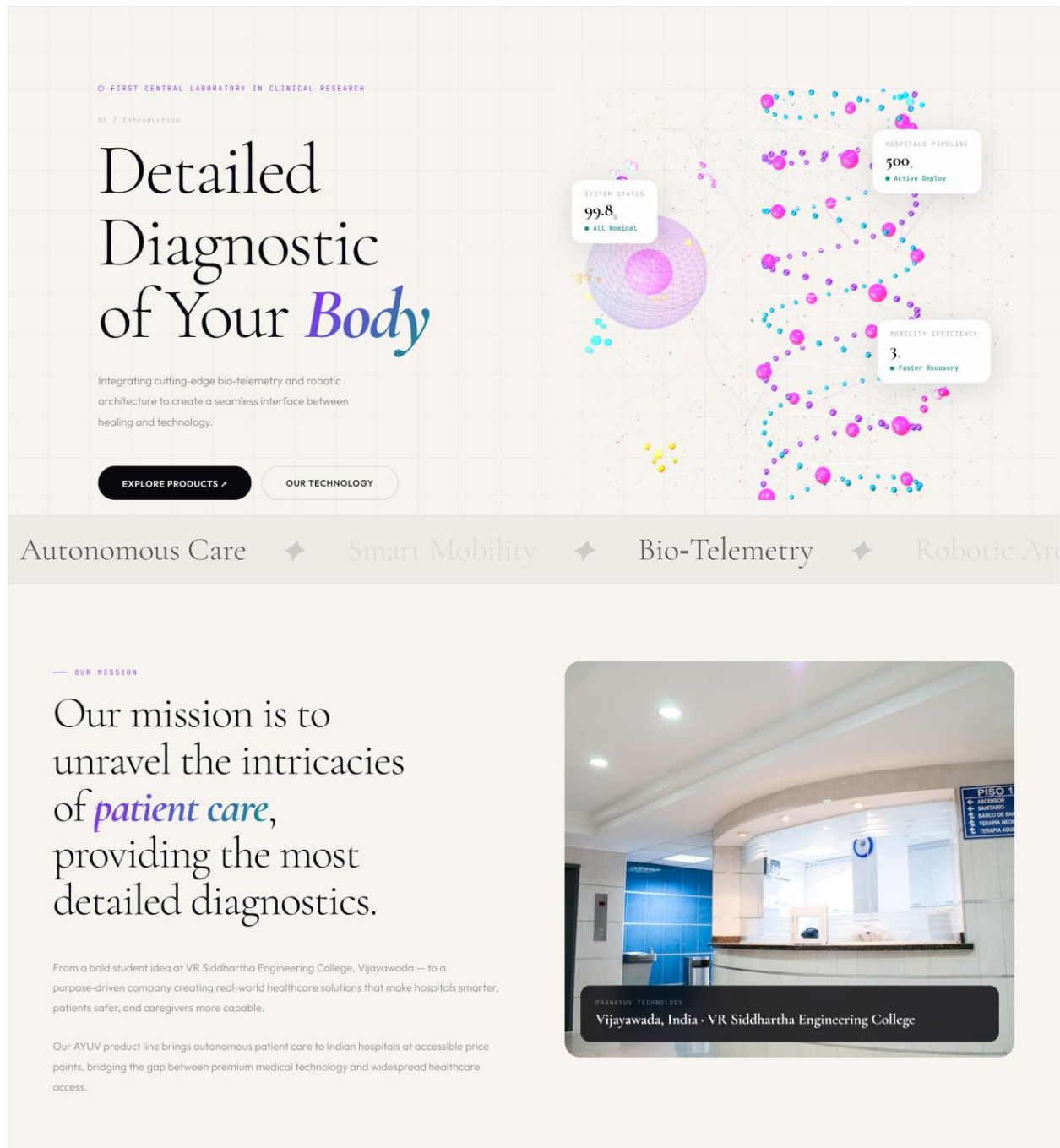


Figure 13.1: Pranayuv Corporate Landing Page featuring custom cursor animation, fluid typography, and dynamic statistics grid.

9. Project 2 – EduNexus School Website Documentation

Project 2 focused on advanced UI/UX design methodology, transforming academic platform requirements into comprehensive wireframes and functional prototypes in Figma, subsequently converting them into clean, responsive HTML5/CSS3 web code within the 'figma/' repository directory.

Design System & Color Psychology

The branding palette centers around an authoritative Olive-Green primary hue (`#6B8E23`) complemented by warm earth tones and crisp white structural canvas spaces. This color psychology was intentionally chosen to foster an academic atmosphere of focus, calm concentration, and trust, contrasting against typical high-stress educational interfaces.

Verified Module Analysis

- **EduNexus Olive Theme Portal (EduNexus-olive-theme.html):** Serves as the central navigation hub featuring unified visual iconography, clear call-to-action enrollment buttons, and card-based academic department highlights.
- **Admission Requirements Gateway (admission-requirements.html):** Deconstructs rigorous admission guidelines into scannable accordion-style requirement cards, document checklists, and interactive timeline steps.
- **Student Dashboard V2 (student-dashboard-v2.html):** A modular, widget-driven interface allowing students to view live subject GPA evaluations, weekly interactive timetable schedules, submission deadlines, and campus notice boards.
- **Principal Executive Dashboard (principal-dashboard.html):** A centralized control panel designed for institutional leaders, showcasing macro KPI metric blocks (total enrollment, attendance percentages, faculty distribution) and rapid broadcast tools.



Figure 14.1: EduNexus Olive Theme Home Page showcasing the custom olive-green design system and card architecture.

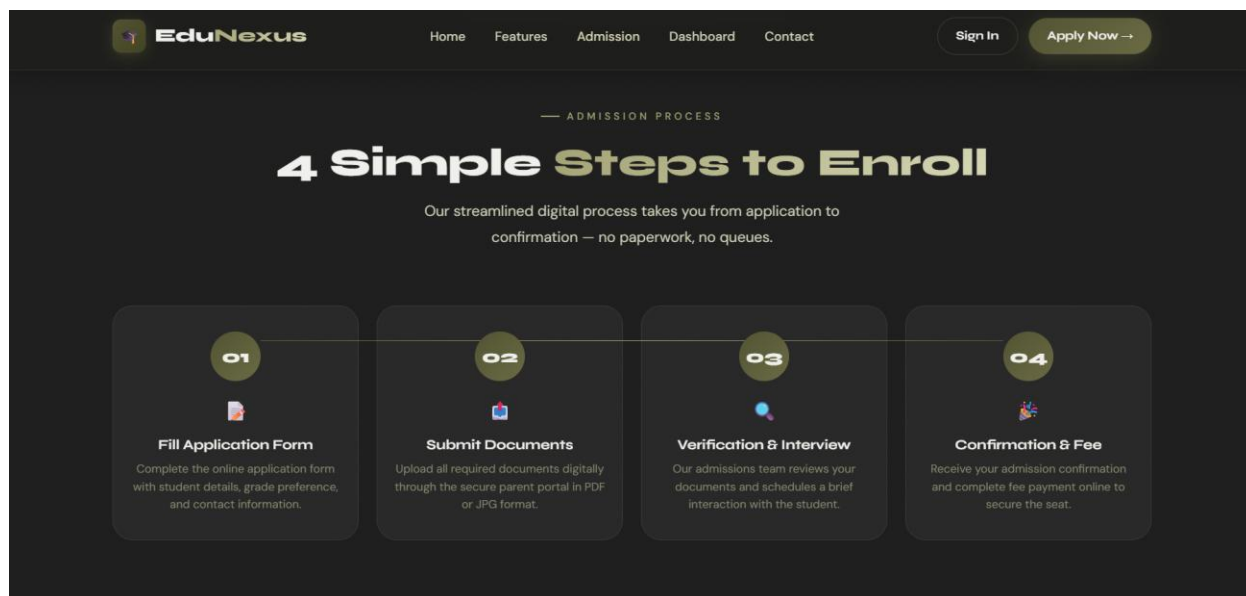


Figure 14.2: Admission Requirements Page illustrating clean typographic hierarchy and multi-step timeline guidance.

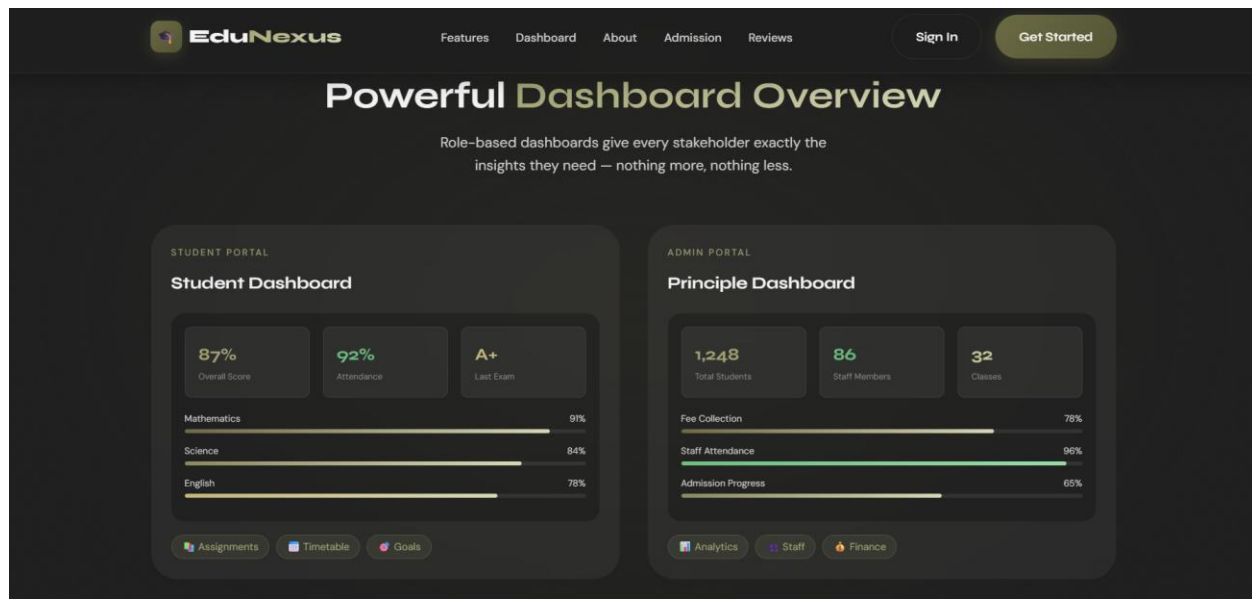


Figure 14.3: EduNexus Student Dashboard and Principle Dashboard displaying academic analytics widgets and responsive schedule grids.

10. Project 3 – BrightPath Academy School Management System

BrightPath Academy (`school\_management/`) represents a comprehensive, multi-module educational management web application designed to digitize school administration and streamline parent-school interaction. Built upon HTML5, custom CSS3, Vanilla JavaScript, and Bootstrap 5.3, the platform integrates robust reactive UI components and data visualization.

Key Verified Technical Features

- **Embedded AI Chatbot Assistance:** An interactive conversational widget embedded into the portal interface (`index.html`). The chatbot assistant simulates automated real-time answers to parent queries regarding fee structures, admission deadlines, uniform policies, and school timings without human administrative intervention.
- **Multi-Step Validated Admission Portal:** Implemented in `admission.html` and `registration.html`, this portal guides guardians through structured enrollment phases (Student Profile, Parent Details, Academic History, Document Attachment). Custom JavaScript event listeners enforce strict real-time field validation before enabling progression.

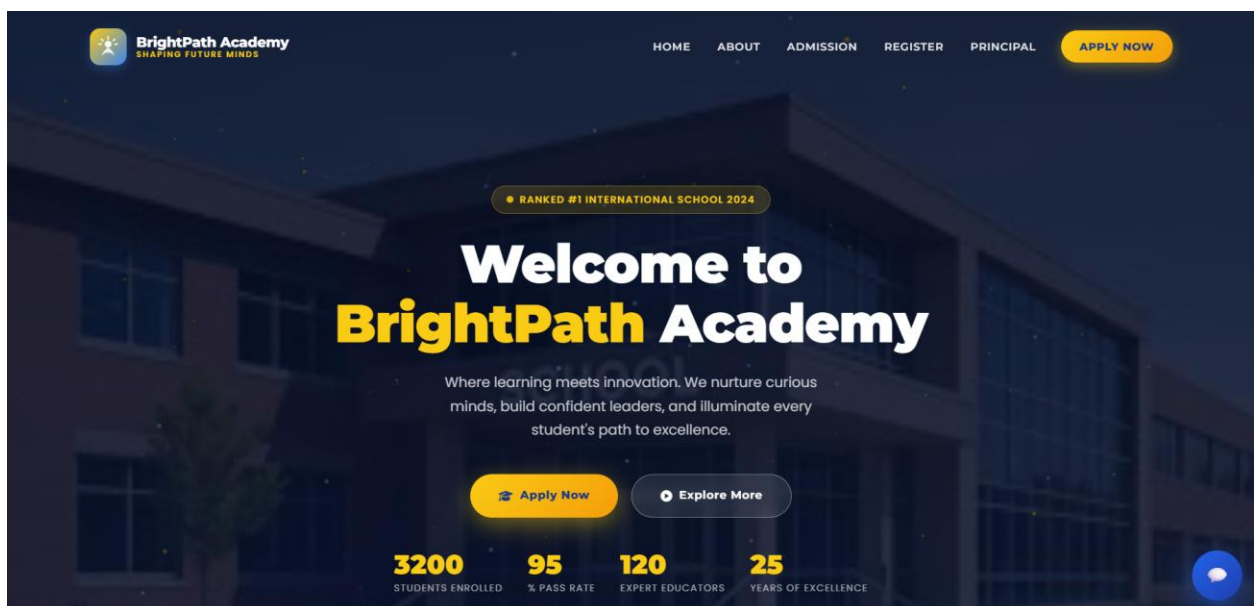


Figure 15.1: BrightPath Academy Home Page featuring welcoming hero graphics, program cards, and interactive AI chatbot entry point.

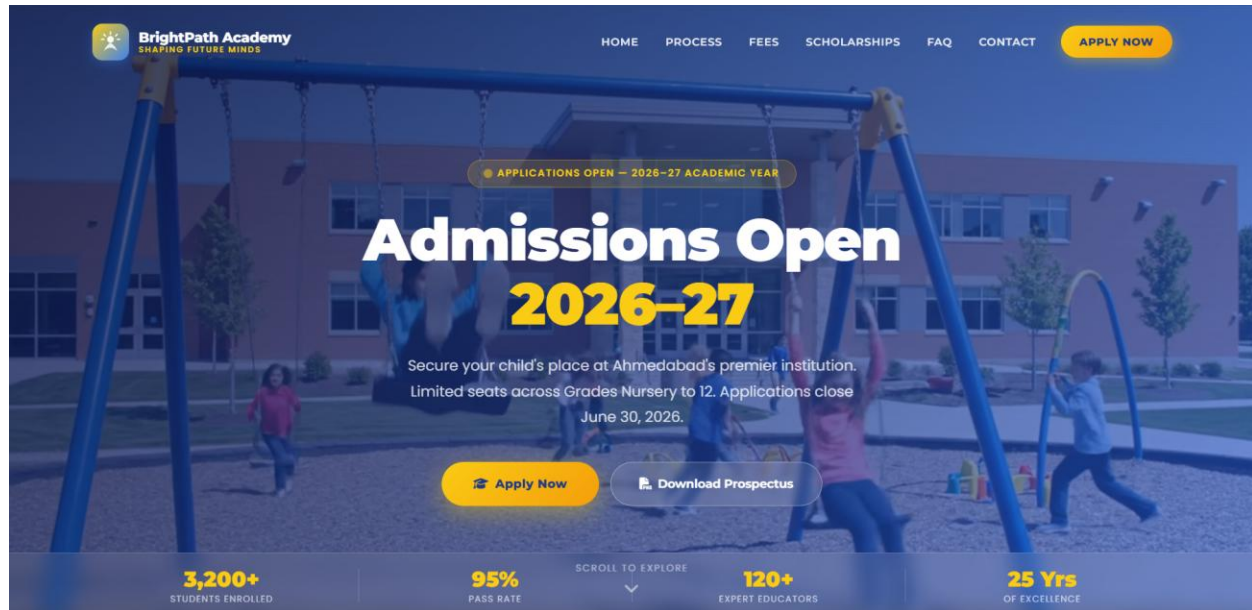


Figure 15.2: BrightPath Academy Multi-Step Admission Portal showing step progress bar and responsive validation inputs.

Institutional Data Visualizations (Chart.js Integration)

To provide actionable educational insights, the Student Dashboard (`student-dashboard.html`) and Principal Dashboard (`principal-dashboard (1).html` & `analysis.html`) leverage the Chart.js library to render dynamic HTML5 canvas telemetry:

- **Student Academic Trending:** Plots historical exam score percentages across academic terms using comparative bar charts and GPA radial progress indicators.
- **Principal Institutional Analytics:** Visualizes monthly attendance compliance curves and school-wide grade averages, allowing school leadership to quickly identify departments or cohorts requiring instructional support.

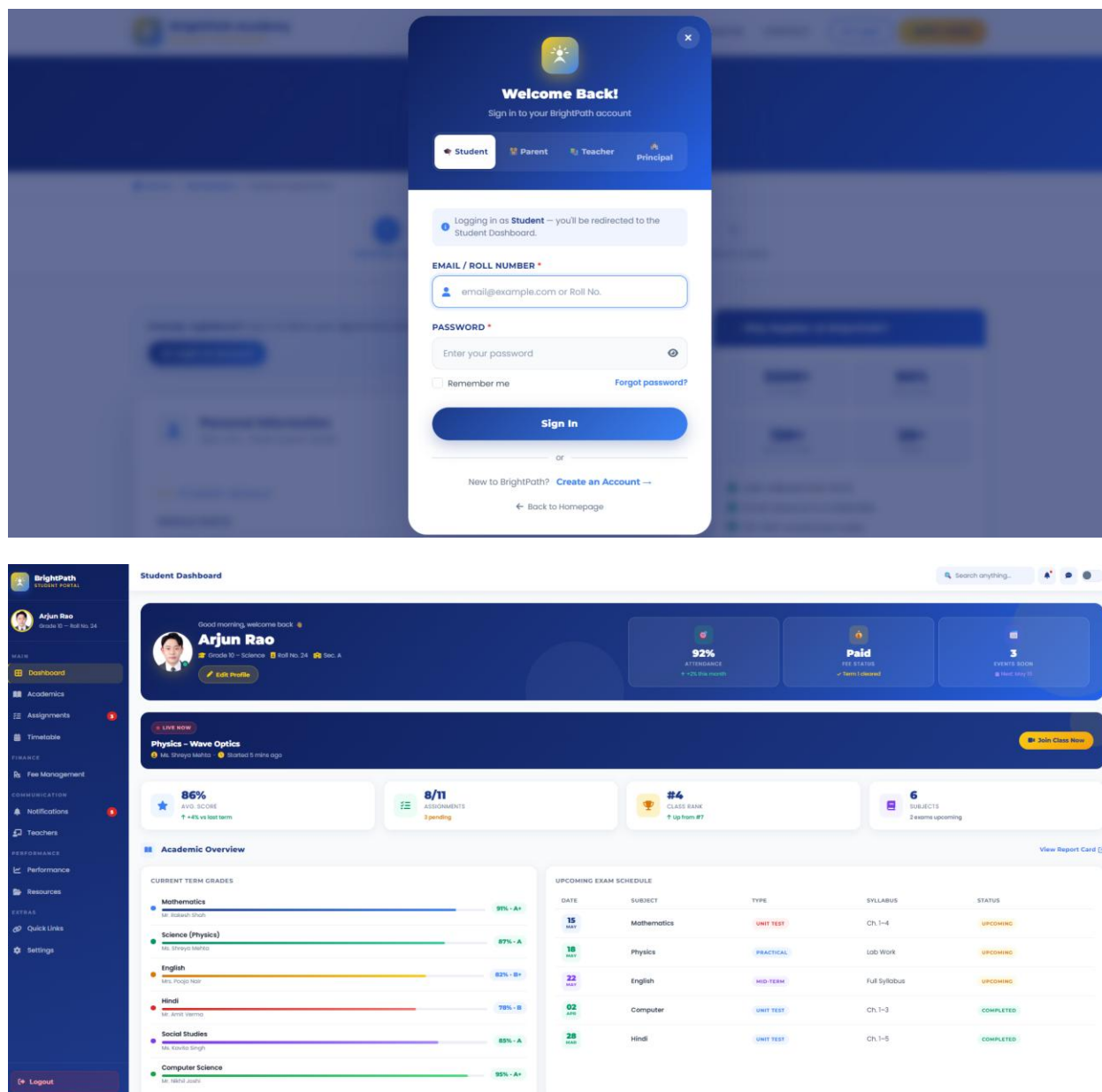


Figure 15.3: BrightPath Academy Student Dashboard featuring real-time Chart.js grade graphing and attendance progress visualization.

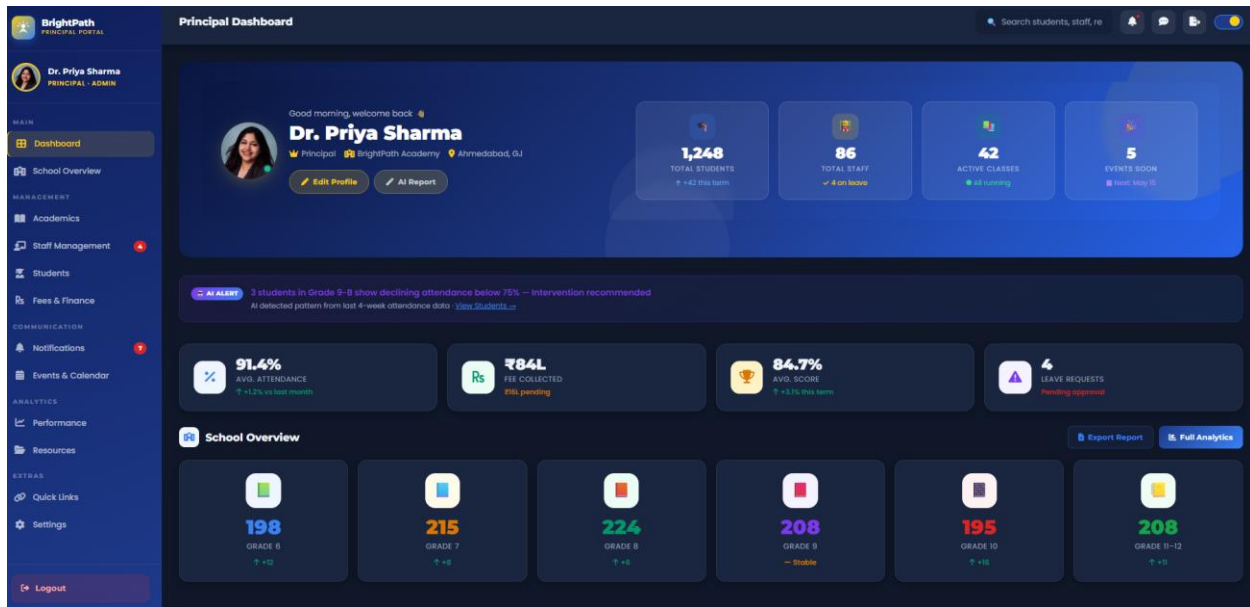


Figure 15.4: BrightPath Academy Principle Dashboard featuring real-time Chart.js grade graphing and attendance progress visualization.

11. Project 4 – Sri Srinivasa Clean Rooms & Medical Furniture Web Application

The Sri Srinivasa Clean Rooms Web Application is the primary full-stack engineering deliverable of this internship. Developed for a specialized medical furniture and operating theater clean room equipment manufacturer based in Hyderabad, Telangana, the software operates simultaneously as an interactive public e-commerce catalog and a secure, centralized corporate administrative platform and CRM.

Architectural Overview

The project adopts an enterprise decoupled Single-Page Application (SPA) architecture. The client frontend is compiled with React 19 and Vite 8, styled using Tailwind CSS v4 and custom CSS variables, and communicated via RESTful JSON fetches to a standalone Node.js/Express.js backend server. Persistent data storage and relational integrity are maintained within a MongoDB database orchestrated via Mongoose schema models.

Verified Public Modules

- **Home Page (Home.jsx):** Creates an authoritative medical brand impression utilizing gradient hero banners, scroll-based navbar backdrop blurring, statistical milestone counters (100+ Products, 50+ Hospitals, 10+ Years), and curated product category highlights (ICU Beds, Hospital Trolleys, Medical Lockers, Clean Room Systems).

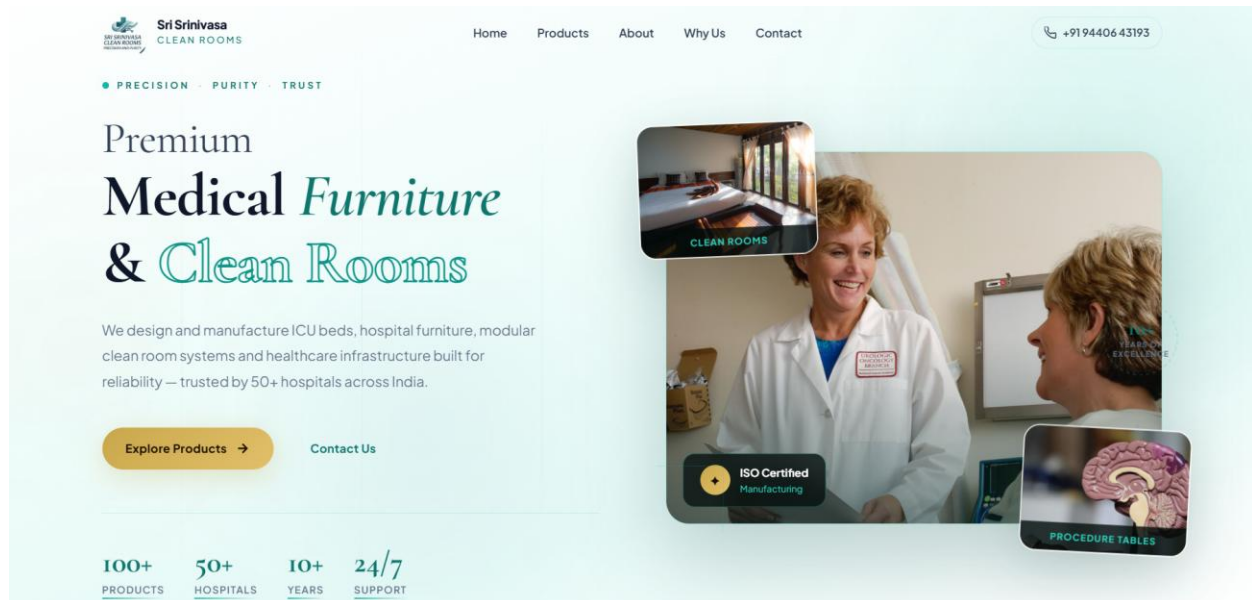


Figure 16.1: Sri Srinivasa Clean Rooms Home Page showcasing the dynamic medical hero section and statistical feature blocks.

- **Product Listing & Catalog Module:** The central product browsing engine ('Products.jsx'). Features instantaneous text searching across title, category, and description fields via debounced React states. Includes database-generated category filter tabs, Low-to-High/High-to-Low price sorting, and paginated grid layout rendering (12 cards per page).
- **Product Detail Specification View:** When selected, individual products render in a deep view ('ProductDetail.jsx') displaying full technical specifications, category classification, and high-resolution zoomable photography.
- **Integrated Customer Quotation Submission Form:** Directly embedded inside 'ProductDetail.jsx', this lead-generation module empowers hospital administrators to submit formal quotation inquiries directly linked to the specific hardware being viewed. Payload inputs include required Name and Phone, optional Email and Message, automatically appending 'productName' and default status flags.

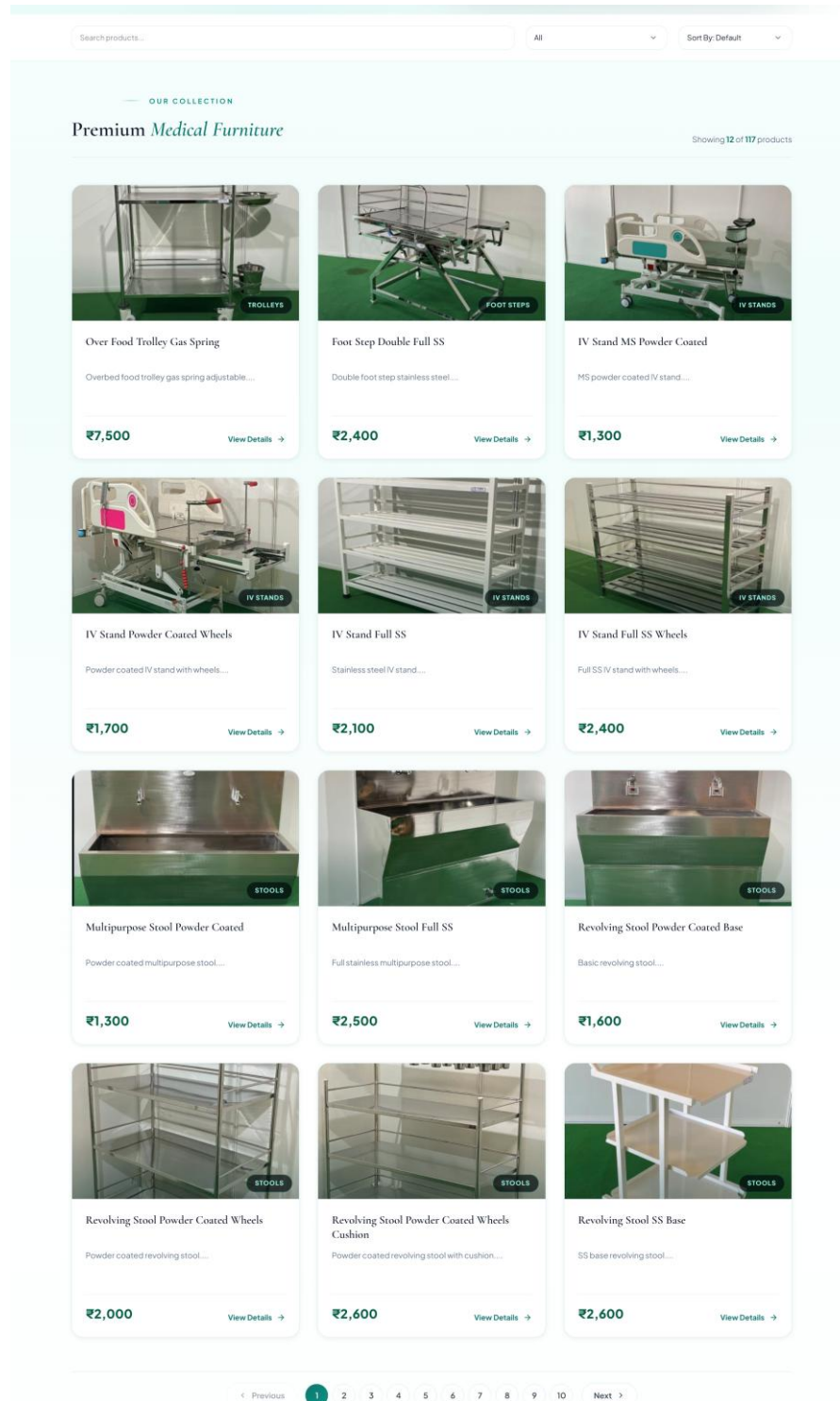


Figure 16.2: Products Page displaying real-time search inputs, dynamic category filter tabs, and responsive medical equipment cards.

Verified Secure Administrative Modules

- **Admin Authentication Gateway:** A protected authentication gateway (`AdminLogin.jsx`) utilizing username/password credential checks against server-side bcrypt hashed strings. Upon verification, the server issues a cryptographically signed JSON Web Token (JWT), which the browser persists in `localStorage` as `adminToken` to authenticate subsequent administrative REST requests.

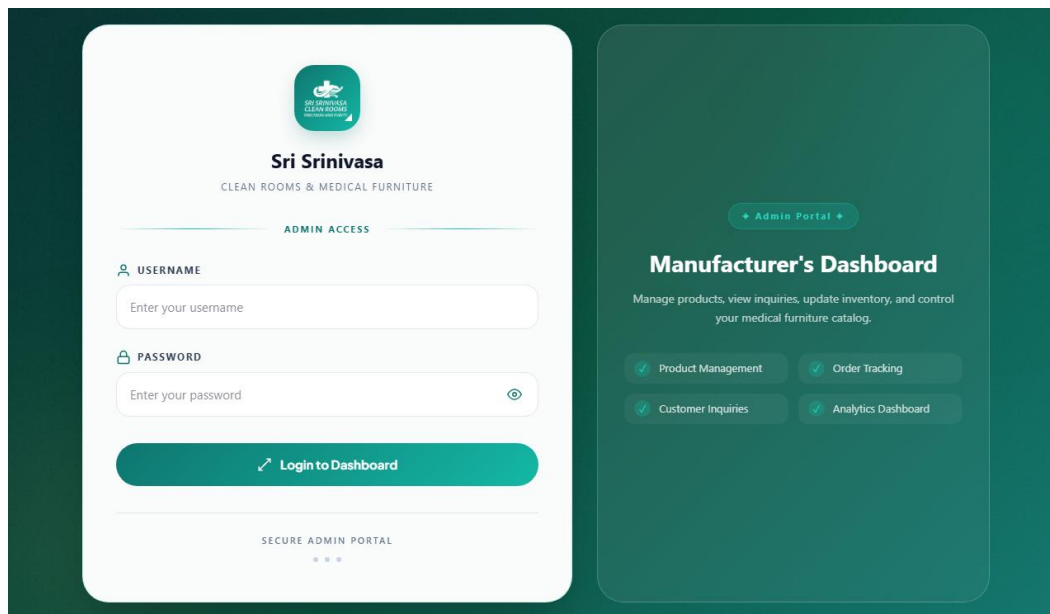


Figure 16.3: Admin Login Portal providing secure JWT authentication and password hashing protection.

- **Admin Dashboard & Product CRUD Interface:** An intuitive, multi-tab administrative interface (`AdminDashboard.jsx`). The Product Management tab allows administrators to perform full Create, Read, Update, and Delete (CRUD) operations. Products are cleanly grouped under collapsible category headers with active count badges.
- **Category Management System:** Allows administrators to dynamically create new equipment categories, modify existing terminology, or safely purge unused classifications with duplicate protection.
- **Inquiry & Lead Management (CRM) Suite:** A functional lightweight CRM system embedded into the admin panel. Administrators can review all incoming customer quote requests in structured table views, filtering across lead priorities (*Normal, Hot Lead, Bulk Order, Dealer, Urgent, Hospital*) and lifecycle workflows (*New, Contacted, Quotation Sent, Negotiation, Closed, Rejected*). Features timestamped internal note taking and one-click deep links to initiate immediate WhatsApp or Email vendor responses.

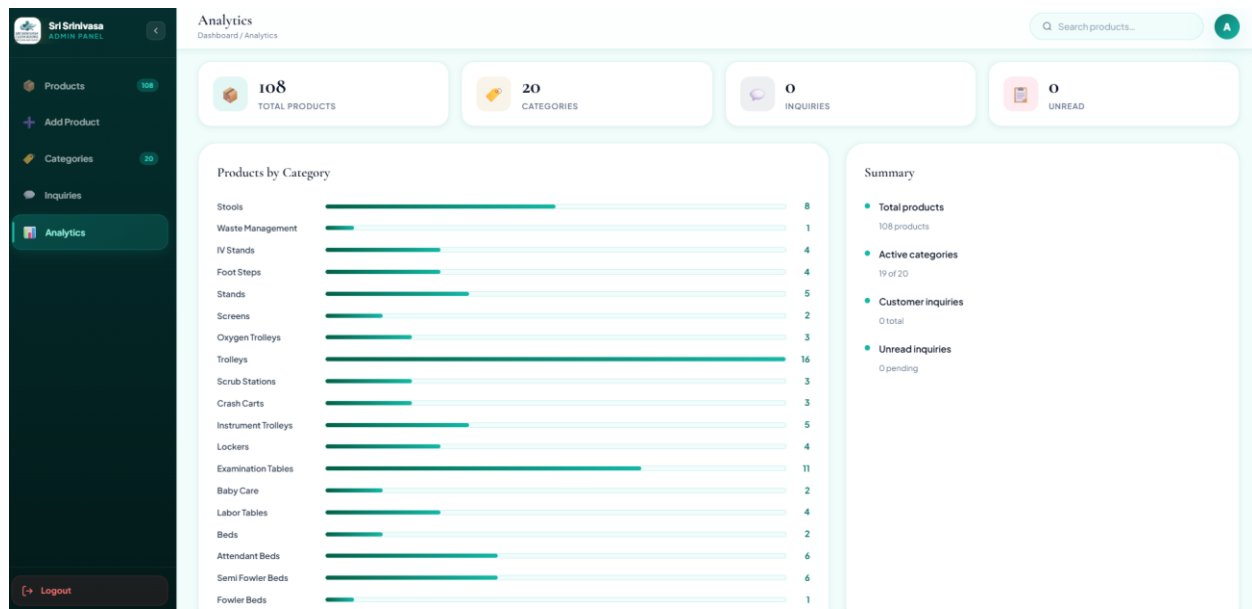


Figure 16.4: Admin Dashboard illustrating centralized category grouped product CRUD operations and lead management controls.

12. Overall Internship Summary

Over the course of the six-month industrial training program (March 13, 2026 to September 13, 2026), approximately 180 hours of rigorous architectural design, full-stack programming, and debugging were invested across four primary web applications and technical research reports. The table below presents a comparative executive summary of the four core projects executed:

Project Identifier & Name	Core Technology Stack	Verified Functional Modules	Engineering Focus & Key Deliverable
Project 1: Pranayuv Landing Page	HTML5, CSS3, Vanilla JavaScript	• Animated Hero Banner • Statistics & HUD Grid • Interactive Custom Cursor	High-end corporate branding website emphasizing canvas rendering animations and responsive fluid layout typography.
Project 2: EduNexus School Website	Figma UI/UX, HTML5, CSS3 Variables	• Olive-Green Theme Home • Admission Requirements • Student Dashboard V2 • Principal Admin Panel	UI/UX design system engineering in Figma translated into pixel-perfect responsive HTML5/CSS3 educational management prototypes.
Project 3: BrightPath Academy Portal	HTML5, CSS3, JS, Bootstrap 5, Chart.js	• Interactive AI Chatbot • Multi-Step Admission Form • Student Analytics Portal • Principal Executive Suite	Frontend institutional portal combining Bootstrap responsive grids, Chart.js visual telemetry, and AI chatbot conversational UX.
Project 4: Sri Srinivasa Clean	React 19, Node.js, Express,	• Live Search Product Catalog • Quote Inquiry Submission	Enterprise full-stack e-commerce web application running Mongoose

Project Identifier & Name	Core Technology Stack	Verified Functional Modules	Engineering Focus & Key Deliverable
Rooms	MongoDB, Multer	• JWT Secure Admin Login • Full CRUD Product Admin • Interactive CRM Lead Panel	schemas, REST APIs, local Multer uploads, and CRM workflows.

Each project incrementally elevated in complexity, culminating in Project 4, which represents a fully realized, production-ready full-stack enterprise web application complete with database security and internal CRM tracking.

13. Skills Learned & Technical Competencies Acquired

The six-month internship training at Pranayuv Technologies bridged the critical gap between academic theoretical coursework and professional software enterprise execution, building robust competencies across modern web engineering domains:

Engineering Domain	Acquired Technical Competence	Practical Codebase Application
Full-Stack Architecture	MERN Stack Development (React, Node, Express, MongoDB)	Engineered and deployed Project 4 from scratch, connecting reactive frontend view engines with robust Express backend servers.
Frontend React Mastery	React 19 Hooks, Custom CSS3, Tailwind utility styling	Mastered reactive state control ('useState', 'useEffect', 'useMemo'), clean SPA routing, and aesthetic UI implementation.
NoSQL Database Design	Mongoose schema modeling, compound indexing, soft deletion	Constructed resilient schemas for products, leads, categories, and audit logs with text-search index optimizations.
Web Security & API Design	JWT session signing, Bcrypt hashing, Joi validation	Protected corporate inventory routes, encrypted admin passwords, and sanitized user incoming data payloads against injection.
UI/UX & Prototyping	Figma wireframing, color psychology, ergonomic layout Design	Designed intuitive, low-stress educational dashboards (EduNexus) and clean medical supply interfaces.

14. Conclusion

The six-month summer internship program (180 hours completed) at Pranayuv Technologies Pvt. Ltd. has been an exceptionally fulfilling and intellectually rewarding culmination of my undergraduate Bachelor of Science in Information Technology studies at Shree Ramkrishna Institute of Computer Education and Applied Sciences, Sarvajanic University.

Throughout this intensive tenure, the foundational objective of developing modern, high-performance web applications and advanced UI/UX design systems was fully realized across four real-world software deliverables. From crafting visually prestigious corporate landing portals with custom Javascript animation algorithms (Project 1), to architecting serene, ergonomic educational management prototypes in Figma (Project 2) and responsive interactive school administration websites with embedded chatbot interfaces (Project 3), every project expanded my versatility as a modern software engineer.

The capstone deliverable—the Sri Srinivasa Clean Rooms Full-Stack E-Commerce & CRM platform (Project 4)—solidified my mastery over the MERN stack architecture. Constructing an application that successfully protects admin authentication via Bcrypt and JWTs, ingests real-time product photography via Multer, queries MongoDB text indexes, and orchestrates corporate sales leads in an integrated CRM panel has equipped me with industry-ready full-stack capabilities.

Ultimately, this internship successfully bridged the gap between academic programming exercises and enterprise-grade software execution. The technical disciplines, defensive programming methodologies, and collaborative communication skills instilled by my corporate mentor, Mr. Ajay Polavarapu, and faculty guides have established a resilient framework for my upcoming career as an advanced UI/UX designer and full-stack software engineer.